Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / Bending
Case	a248dad6-5ed8-4dae-a374-cb93962d61eb
Date	2026-11-07 07:48:48
Status	Reviewed by expert (expert)

WARNING: the AI model was not used for this analysis (fake-offline). The text below is a generic offline template, not a result.

1. Problem

Defect	Severity	Confidence	Detail
scratch	minor	92%	
dent	major	88%	
crack	critical	76%	
deformation	major	91%	

- L2: -0.11 mm
- H1: -0.05 mm

2. Root cause

Cause: Improper Bending Force / Over Bending

Reason: The detected deformation and dimensional inaccuracy strongly suggest excessive or improperly calibrated bending force. Retrieved RCA documents confirm this pattern for SPCC steel bending operations.

Effect: Deformation, Dimensional Inaccuracy

Confidence: 68%

Supporting evidence
- **seed-001**: Improper Bending Force or Over Bending causes deformation and dimensional inaccuracy. Precautions: optimize bending parameters, use proper tooling, calibrate machines regularly.
- **seed-004**: Springback in sheet metal bending causes dimensional inaccuracy. Overbend compensation and material property testing are required.

3. Solution

Corrective actions (fix this occurrence)

1. Fix A

Preventive actions (stop it recurring)

1. Prevent B

Precautions

- Optimize bending parameters to material specification
- Use proper tooling and punch/die sets
- Regular machine calibration and force verification

How to verify the fix

Re-inspect the next 30 parts; no deformation and all dimensions within tolerance

4. Other causes considered

Rank	Cause	Confidence	Reason
2	Material Inconsistency / Hardness Variation	65%	Hardness variation in the incoming material can cause unpredictable springback and cracking. Material certificate verification is required.
3	Tool Wear / Die Damage	62%	Progressive die wear causes scratches, dents, and edge burrs. Preventive maintenance schedule should be implemented.

5. Traceability

AI model: fake-offline. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.